Nurture Course

PHASE : SPEED & ACCURACY TEST
TARGET : PRE-MEDICAL 2023

Test Type :**PRACTICE TEST** Test Pattern :**NEET (UG)**

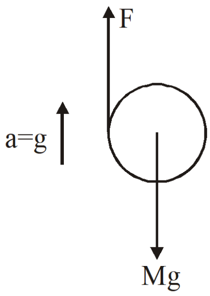
TEST DATE :

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	4	4	4	1	2	3	2	3	1	1	2	3	1	1	2	4	3	3	3	2	2	1	4	4	3	2	3	3	3	4
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	1	1	2	1	2	4	4	2	3	1	4	2	3	2	3															

HINT – SHEET

1. Ans (4)



$F - Mg = Mg \Rightarrow F = 2\,Mg$

$FR = I\alpha = \dfrac{MR^2}{2}\alpha$

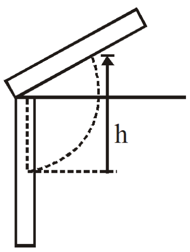
$\Rightarrow \quad F = \dfrac{M}{2}(R\alpha) \quad \Rightarrow \quad \dfrac{2(2Mg)}{M} = R\alpha$

$\Rightarrow \quad R\alpha = 4g$

2. Ans (4)

$TE_i = TE_f$

$\dfrac{1}{2}I\omega^2 = mgh$

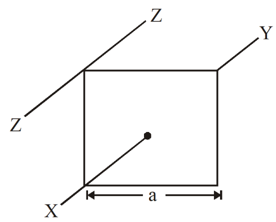


$\dfrac{1}{2} \times \dfrac{1}{3}m\ell^2\omega^2 = mgh$

or $\quad h = \dfrac{1}{6}\dfrac{\ell^2\omega^2}{g}$

3. Ans (4)

Moment of inertia of the square plate about XY is $\dfrac{ma^2}{6}$ moment of inertia about ZZ can be computed using parallel axis theorem,



$I_{ZZ} = I_{XY} + m\left(\dfrac{a}{\sqrt{2}}\right)^2$

$= \dfrac{ma^2}{6} + \dfrac{ma^2}{2} = \dfrac{2ma^2}{3}$

4. Ans (1)

According to theorem of parallel axes,

$I = \dfrac{2}{5}M\left(\dfrac{R}{2}\right)^2 + M(2R)^2 + \dfrac{2}{5}M\left(\dfrac{R}{2}\right)^2$

$= 4MR^2 + \dfrac{1}{5}MR^2 = \dfrac{21}{5}MR^2$

5. Ans (2)

$I = MR^2 + 3\left(\dfrac{mR^2}{3}\right) = (M+m)R^2$

6. Ans (3)

$I = m\left(\dfrac{r}{2}\right)^2 + m\left(\dfrac{r}{2}\right)^2 = \dfrac{mr^2}{2}$

7. Ans (2)

When a wire is bent in the form of a circular ring, then radius of the wire,

$r = \frac{\ell}{2\pi}$

Moment of inertia of the ring about its axis,

$I = mr^2 = m\left(\frac{\ell}{2\pi}\right)^2 = \frac{m\ell^2}{4\pi^2}$

8. Ans (3)

M = Mass of the square plate before cutting the holes Mass of one hole,

$m = \left(\frac{M}{16R^2}\right)\pi R^2 = \frac{\pi M}{16}$

$\therefore$ Moment of inertia of the remaining portion,

$I = I_{square} - 4I_{hole}$

$= \frac{M}{12}(16R^2 + 16R^2) - 4\left[\frac{mR^2}{2} + m(2R^2)\right]$

$= \frac{8}{3}MR^2 - 10mR^2$

$= \left(\frac{8}{3} - \frac{10\pi}{16}\right)MR^2$

9. Ans (1)

Moment of inertia of solid sphere of mass M and radius R about an axis passing through the centre of mass is : $I = \frac{2}{5}MR^2$. Let the radius of the disc is r.

Moment of inertia of circular disc of radius r and mass M about an axis passing through the centre of mass and perpendicular to its plane $= \frac{1}{2}Mr^2$.

Using theorem of parallel axes, moment of inertia of disc about its edge is:

$I' = \frac{1}{2}Mr^2 + Mr^2 = \frac{3}{2}Mr^2$

Given : $I = I'$

or $\frac{2}{5}MR^2 = \frac{3}{2}Mr^2$

or $r^2 = \frac{4}{15}R^2$

or $r = \frac{2R}{\sqrt{15}}$

10. Ans (1)

If mass 'm' and radius 'r', then

ring $\Rightarrow I_{XX'} = \frac{3}{2}mr^2$

Here $m = r\ell$, $\ell = 2\pi r$

$\therefore I_{XX'} = \frac{3\rho\ell^3}{8\pi^2}$

11. Ans (2)

Moment of 4 N force

$= 4 \times \frac{20}{100}$ N-m (anti-clockwise)

Moment of 8 N force

$= 8 \times \frac{20}{100} \times \sin 30°$ N-m (clockwise)

Moment of 9 N force $= 9 \times \frac{20}{100}$ N-m (clockwise)

Moment of 6 N force $= 6 \times \frac{20}{100} \times \sin 0° = 0$

$\therefore \tau = \left(4 \times 0.2 - 8 \times 0.2 \times \frac{1}{2} - 9 \times 0.2\right) = -1.8$N-m

= 1.8 N-m(clockwise)

12. Ans (3)

Beam is not at rotational equilibrium, so force exerted by the rod (beam) decrcase

13. Ans (1)

$\vec{\tau} = \frac{d\vec{L}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = m\frac{d}{dt}(\vec{r} \times \vec{v})$

$\vec{\tau} = [16\,t]\,\hat{k}$

at t = 2 $\vec{\tau}_{t=2} = (16 \times 2)\hat{k} = 32\hat{k}$ N-m

14. Ans (1)

$\alpha = \frac{\tau}{I} = 15$ rad/s^2

$\theta = w_0 t + \frac{1}{2}\alpha t^2$

$= \frac{1}{2} \times 15 \times 100 = 750$ rad

15. Ans (2)

$(mg\sin\theta)\,R = \left(\frac{2}{5}mR^2 + mR^2\right)\alpha$

$\Rightarrow \alpha = \frac{5}{7}\frac{g\sin\theta}{R}$, for pure rolling, $a = \alpha R$

by newton's second law,

$mg\sin\theta - \mu mg\cos\theta = ma$

$\Rightarrow \sin\theta - \mu\cos\theta = \frac{5}{7}\sin\theta$

$\Rightarrow \boxed{\mu = \frac{2}{7}\tan\theta}$

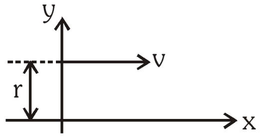
16. Ans (4)

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

$mgh = \frac{1}{2}mv^2 + \frac{1}{2}\frac{MR^2}{2} \times \frac{v^2}{R^2}$

$v = \sqrt{\frac{4mgh}{2m + M}}$

17. Ans (3)



$L = mvr$

18. Ans (3)

$\vec{L} = m(\vec{r} \times \vec{v}) = -10\hat{i} - 15\hat{j} - 12\hat{k}$

19. Ans (3)

$$W = \tau\theta = 8\left(\frac{1}{2} \times \frac{8}{30} \times 4^2\right) = 17.1 \text{ J}$$

20. Ans (2)

$L = mvh$

$$= \frac{m\,(v\cos 45)\;\;(v^2\sin^2 45)}{2g}$$

$$= \frac{mv^3}{4\sqrt{2}g}$$

21. Ans (2)

$\Delta PE + \Delta KE = 0$

$$-mgh + \frac{1}{2}\left(\frac{3}{2}mR^2\right)\frac{v^2}{R^2} = 0$$

$$v = \sqrt{\frac{4}{3}gh}$$

22. Ans (1)

Required fraction

$$= \frac{K_R}{K_R + K_T} = \frac{\frac{1}{2}I\omega^2}{\frac{1}{2}I\omega^2 + \frac{1}{2}Mv^2}$$

$$= \frac{\frac{1}{2}MR^2\omega^2}{\frac{1}{2}MR^2\omega^2 + \frac{1}{2}Mv^2}$$

$$= \frac{MR^2\,(v^2/R^2)}{MR^2\,(v^2/R^2) + Mv^2}$$

$$= \frac{Mv^2}{Mv^2 + Mv^2} = \frac{1}{2}$$

23. Ans (4)

no rolling on smooth incline, so all will take equal time.

24. Ans (4)

$$\frac{1}{2}mv^2\left[1 + \frac{k^2}{R^2}\right] = \frac{1}{2}kx^2$$

$$\frac{1}{2} \times 3 \times (4)^2\left[1 + \frac{1}{2}\right] = \frac{1}{2} \times 200\;x^2$$

$$\Rightarrow x^2 = \frac{3 \times 4 \times 4 \times 3}{400} \Rightarrow x = \frac{3 \times 4}{20} = 0.6\text{m}$$

25. Ans (3)

$$KE = \frac{1}{2}I\omega^2 + \frac{1}{2}Mv_{cm}^2$$

$$I = \frac{MR^2}{2},\quad v_{cm} = \omega R$$

$$KE = \frac{1}{2}I\omega^2 + \frac{1}{2}M\omega^2R^2$$

$$= \frac{1}{2}I\omega^2 + \frac{2}{2}I\omega^2$$

$$= \frac{3}{2}I\omega^2$$

26. Ans (2)

$v = \sqrt{\dfrac{2gh}{1 + \frac{k^2}{r^2}}}$ solid sphere reaches with larger speed

since potential energy converted into kinetic energy so kinetic energy of both at bottom will be same since solid sphere has larger velocity so its momentum will be larger.

27. Ans (3)

$\omega R = 2$

$\therefore \omega(2R) - 0 = 2\omega R = 4$ m/s

28. Ans (3)

$$KE = \frac{L^2}{2I} = \frac{(I_1\omega_1 + I_2\omega_2)^2}{2(I_1 + I_2)}$$

29. Ans (3)

$(I_1 + I_2)\omega' = I_1\omega_1 + I_2\omega_2$

$$\omega' = \frac{\frac{MR^2}{2}\omega}{\frac{MR^2}{2} + \frac{1}{2}\frac{M}{u}R^2} = \frac{4}{5}\omega$$

30. Ans (4)

$L = I\omega$

$\omega' = 2\omega$

$$\frac{1}{2}\left(\frac{1}{2}I\omega^2\right) = \frac{1}{2}I'\omega'^2$$

$$\frac{I\omega^2}{2} = I'\,4\omega^2$$

$$I' = \left(\frac{I}{8}\right)$$

$$L' = I'\,\omega' = \frac{I}{8}2\omega = \frac{I\omega}{4} = \left({}^{L}\!/_{4}\right)$$

31. Ans (1)

$I_1\omega_1 = I_2\omega_2$

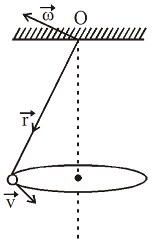
$$mR^2\omega = \frac{mR^2}{4}\omega_2$$

$\omega_2 = 4\omega$

$$KE = \frac{1}{2}mR^2\omega^2$$

$$\frac{KE_1}{KE_2} = \frac{R_1^2\omega_1^2}{R_2^2\omega_2^2} = 4$$

32. **Ans (1)**



The direction of $\vec{\omega}$ vector is always $\perp^r$ to the plane containing $\vec{r}$ & $\vec{v}$, as shown in fig. (at any instant).

Angular momentum

$\vec{L} = I\vec{\omega}$

Obviously, $\vec{L}$ changes, as $\vec{\omega}$ changes in direction but $\vec{\tau} \| d\vec{\theta}$.

$\therefore$ No work is done by torque due to weight of bob.

So $|\vec{\omega}| =$ constant or $|\vec{L}| =$ constant

33. **Ans (2)**

Here, $u = V_0$, $\omega_0 = -\dfrac{V_0}{2R}$

At pure rolling ;

$$V = V_0 - \left(\frac{F_f}{m}\right)t \;\&\; \frac{V}{R} = -\frac{V_0}{2R} + \left(\frac{F_f}{m.R}\right)t$$

(In pure rolling $V = R\omega$)

$(\alpha = \dfrac{\tau}{I} = \dfrac{F_f . R}{mR^2})$

$$\Rightarrow V_0 - V = V + \frac{V_0}{2} \Rightarrow 2V = \frac{V_0}{2} \Rightarrow V = \frac{V_0}{4}$$

34. **Ans (1)**

$\tau = I\,\alpha$
$\tau_A = \tau_B$
$I_A\,\alpha_A = I_B\,\alpha_B$
$I_A < I_B$
$\alpha_A > \alpha_B$
$\omega_A > \omega_B$

35. **Ans (2)**

$L = (\vec{r} \times m\vec{v}) = -\,mvb\,\hat{k}$

36. **Ans (4)**

$$\vec{L} = \vec{r} \times \vec{p} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -1 \\ 3 & 4 & -2 \end{vmatrix}$$

$$\vec{L} = \vec{r} \times \vec{p} = \hat{i}(-4+4) - \hat{j}(-2+3) + \hat{k}(4-6)$$

$= 0\hat{i} - 1\hat{j} - 2\hat{k}$ $\vec{L}$ has components along –y-axis and -z-axis but it has no components in the x-axis. The angular momentum is in Y–Z plane, i.e. perpendicular to x-axis.

37. **Ans (4)**

$\theta = 2t^3 - 6t^2$
$\omega = 6t^2 - 12t$
$\alpha = 12t - 12 = 0$
$\alpha = 0 \rightarrow t = 1$ sec.

38. **Ans (2)**

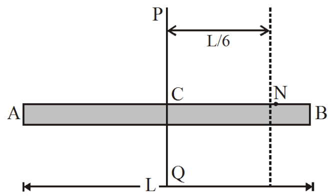
Power, $P = \tau \times \omega$

so, $\tau = \dfrac{P}{\omega} = \dfrac{1200}{600} = 2$ N-m

39. **Ans (3)**

Moment of inertia of the rod about a perpendicular axis PQ passing through the centre of the mass C,

$$I_{CM} = \frac{ML^2}{12}$$



Thus, the moment of inertia I' about an axis parallel to PQ and passing through the point N.

$$I' = I_{CM} + M\left(\frac{L}{6}\right)^2 = \frac{ML^2}{12} + \frac{ML^2}{36} = \frac{ML^2}{9}$$

If K be the radius of gyration, then

$$K = \sqrt{\frac{I'}{M}} = \sqrt{\frac{L^2}{9}} = \frac{L}{3}.$$

40. **Ans (1)**

$P = \tau\,.\,\omega$

$$P = I\,\alpha\,.\,\omega = I\left(\omega\frac{d\omega}{d\theta}\right).\,\omega$$

$\omega^2 d\omega = d\theta$
$\omega \propto \theta^{1/3}$
$\omega \propto (n)_{1/3}$

42. **Ans (2)**

$I = (5(0.2)^2 + 2(0.4^2) \times 2 = 1.04$ kg m^2

44. **Ans (2)**

According to theorem of parallel axis,

$I = I_{CG} + M(2R)^2$

where, I_{CG} = MI about an axis through the centre of gravity.

$$I = \frac{2}{5}MR^2 + 4MR^2 = \frac{22}{5}MR^2$$

or $\quad MK^2 = \dfrac{22}{5}MR^2$

$\therefore \quad K = \sqrt{\dfrac{22}{5}}R$